Module II: Styling Pages

LaTeX Made Simple · Syllabus units 5–7 · Primer chapters 6–7 (pages 55–94)

6. Displayed text and lists

SYLLABUS UNIT 5 · PRIMER PAGES 55–68

> **In plain words.** "Displaying" means setting a passage apart from the main text, for example an indented quotation, a poem, or a list. LaTeX does this with *environments*: a `\begin{name}` and an `\end{name}` around the passage.

6.1 Quotations

ENVIRONMENT	USE	LOOK
`quote`	A quotation that is one paragraph	Indented on both sides
`quotation`	A quotation of several paragraphs	Indented on both sides, and the first line of each paragraph is indented too

```
\begin{quote}
  The true spirit of delight ... from which all great work springs.
\end{quote}
```

Paragraphs inside `quotation` are separated by blank lines, as in ordinary text.

6.2 Poetry: verse

```
\begin{verse}
A mathematician once confided\\
That a M\"obius band is one sided\\
You'll get quite a laugh\\
If you cut it in half\\
For it stays in one piece when divided
\end{verse}
```

- `\\` forces a line break. Stanzas are separated by a blank line.
- `\\*` forces the break but forbids a page break there, so a rhyming couplet stays together.
- `\\[5pt]` adds 5 points of extra space after the line.

6.3 Lists

> **In plain words.** LaTeX has three list environments. Each item is marked with `\item`. Lists can be nested up to four levels deep.

ENVIRONMENT	LABELS	USE FOR
`itemize`	Bullets: •, –, *, · at the four levels	Unordered points
`enumerate`	Numbers: 1., (a), i., A.	Steps, or points you refer to by number
`description`	Whatever you give in square brackets after `\item`, in bold	Terms and their meanings

BULLETS

```
One should keep the following in mind when using \TeX
\begin{itemize}
\item \TeX\ is a typesetting language and not a word processor
\item \TeX\ is a program and not an application
\end{itemize}
Being a program, \TeX\ offers a high degree of flexibility.
```

A list inside a list is nested. The second level uses dashes, the third an asterisk, the fourth a dot.

CHANGING THE BULLET

```
{\renewcommand{\labelitemi}{$\triangleright$}
\begin{itemize}
\item First item of a new list
\item Second item
\end{itemize}}
```

The braces keep the change local to this list. The first-level bullet is `\labelitemi`, the second-level `\labelitemii`, and so on. Remove the braces to change the bullet for the whole document.

NUMBERED LISTS

```
\begin{enumerate}
  \item Prepare a source file with the extension "tex"
  \item Compile it with \LaTeX to produce a "dvi" file
  \item Print the document using a "dvi" driver
\end{enumerate}
```

The numbering is 1, 2, 3 at the first level, (a), (b) at the second, i, ii at the third, and A, B at the fourth.

CUSTOM LABELS WITH THE ENUMERATE PACKAGE

```
\usepackage{enumerate}   % in the preamble, by David Carlisle
\begin{enumerate}[\hspace{0.5cm}Step 1.]
\item Prepare a source file with the extension "tex"
\item Compile it with \LaTeX to produce a "dvi" file
  \begin{enumerate}[i.]
  \item Use a previewer to view the output
  \item Edit the source if needed
  \end{enumerate}
\item Print the document using a "dvi" driver
\end{enumerate}
```

The text in square brackets after `\begin{enumerate}` becomes the label of the first item. The `\hspace` adds an indent. The letters `A`, `a`, `I`, `i` and `1` inside the brackets are read as numbering styles. To keep one as plain text, wrap it in braces, as in `[\hspace{1cm}({A}1)]`. The primer's example then shows (A1), (A2), (A3), (A4).

Pitfall (from the primer). Writing `\begin{enumerate}[\hspace{1cm}(A1)]` without braces gives (11), (22), (33). The letter A is read as an instruction to number with capitals. Braces around the A fix it.

OVERRIDING ONE LABEL

```
\begin{enumerate}
  \item[(1)] $l$ is an upper bound of $A$
  \item[(2)] if $u$ is an upper bound of $A$, then $l\le u$
\end{enumerate}
```

Any list can use a custom label in square brackets right after `\item`.

Pitfall: square brackets. Write `\item (0,1) is an open interval` and it works. But `\item [0,1] is a closed interval` loses the `[0,1]`, because LaTeX reads it as a label. Put the maths in dollar signs: `\item $[0,1]$ is a closed interval`.

DESCRIPTIONS

```
\begin{description}
  \item[\TeX] A typesetting program
  \item[Emacs] A text editor and also
  \item[AbiWord] A word processor
\end{description}
```

The label is whatever is in the brackets, typeset in bold roman. There is no indent at the first level by default.

Remember (Part 6)

- Use `quote` for one paragraph and `quotation` for several.
- Inside verse, the line break is `\\`. A blank line starts a new stanza.
- Lists nest up to four levels. Each item starts with `\item`.
- Braces keep a change of label local. Put maths and brackets in `$...$` inside an item.

7. Rows and columns

SYLLABUS UNITS 6 AND 7 · PRIMER PAGES 69–94

In plain words. A list gives you rows. When the rows need columns that line up, use `tabbing` (for simple aligned text) or `tabular` (for a real table with rules and cells). `tabular` is the one you will use most.

7.1 Tabbing: setting tab stops

In `tabbing`, you set tab stops with `\=` and jump to them with `\>`. A line ends with `\\`.

```
\begin{tabbing}
\hspace{1cm}\= \textbf{AbiWord}\quad\= word processor\\[5pt]
            \> \textbf{Emacs}                  \> text editor\\[5pt]
            \> \textbf{\TeX}                   \> typesetting program
\end{tabbing}
```

Line by line: the first `\=` sets a tab 1 cm from the margin; the second sets a tab after the word "AbiWord" plus a quad (a space the width of the letter M). Every later `\>` goes to those tab stops.

Pitfall. A tab is set by the first line that uses `\=`. If that line is wide (for example the word "TEX" above), the later lines jump too far. The fix is a **dummy line** that sets the tabs and is removed with `\kill`:

```
\begin{tabbing}
  \textbf{AbiWord}\quad\= A word processor\kill
  \textbf{\TeX}\quad \> A typesetting program\\[5pt]
  \textbf{Emacs}\quad \> A text editor\\[5pt]
  \textbf{AbiWord}\quad\> A word processor
\end{tabbing}
```

The line ending in `\kill` sets the tab stops but is not printed.

OTHER TABBING COMMANDS

COMMAND	MEANING
`\=`	Set a new tab stop here
`\>`	Jump to the next tab stop
`\\`	End of line
`\kill`	End of line, but do not print it (for setting tabs)
`\+`, `\-`	`\+` at the end of a line makes the following lines start at the first tab; `\+\+` starts them at the second tab. `\-` cancels one `\+`
`\``	Text after it is pushed flush right
`\pushtabs` ... `\poptabs`	Save the tab settings, use new ones, then restore the saved ones

Tabbing cannot be nested, and it can run across pages. Tabular cannot run across pages but can sit inside maths and other tables.

7.2 Tabular: the table environment

> **In plain words.** A `tabular` is a grid. The argument in braces after `\begin{tabular}` gives one letter per column: `l` left, `c` centred, `r` right. Cells are separated by `&`, and each row ends with `\\`.

```
\begin{center}
\begin{tabular}{lr}
Planet &amp; Diameter (km)\\[5pt]
Mercury &amp; 4878\\
Venus   &amp; 12104\\
Earth   &amp; 12756
\end{tabular}
\end{center}
```

The `center` environment centres the table on the page. The `\\[5pt]` adds a little space after the heading row.

RESULT (PLANET TABLE)

PLANET	DIAMETER (KM)
Mercury	4878
Venus	12104
Earth	12756

COLUMN LETTERS

- `l`, `c`, `r`: left, centre, right
- `p{6cm}`: a paragraph cell 6 cm wide, wrapping its text
- `|`: a vertical line

RULES: `\HLINE` AND `\CLINE`

```
\begin{tabular}{|l|r|}
\hline
Planet &amp; Diameter(km)\\
\hline
Mercury &amp; 4878\\
\hline
\end{tabular}
```

- `|` in the column letters draws a vertical line. Two bars give a double line.
- `\hline` draws a horizontal line across the whole table. It must come at the start of a row or right after `\\`.
- `\cline{i-j}` draws a line only under columns i to j.
- Because a final `\hline` follows the last row, that last row still needs its `\\`.

SPANNING COLUMNS: `\MULTICOLUMN`

```
\begin{tabular}{lrr}
Planet &amp; \multicolumn{2}{c}{Distance from sun (km)}\\
       &amp; Maximum   &amp; Minimum\\
Mercury &amp; 69400000 &amp; 46800000\\
\end{tabular}
```

`\multicolumn{2}{c}{text}` makes one cell that spans two columns, centred. The general form is `\multicolumn{num}{pos}{item}`. The empty first cell of the second row begins with `&`, because the row has no entry in column one.

Careful with the last column. When a `\multicolumn` covers the last column and you want a vertical line on its right, put the line in its position argument: `\multicolumn{2}{c|}{...}`. The primer shows that the final `|` is reset by `\multicolumn` and must be repeated.

SPACING: `\ARRAYSTRETCH` AND `@{}`

```
\renewcommand{\arraystretch}{1.2}   % 20% more space between rows
\begin{tabular}{|c|r@{--}l|}          % a dash between the two numbers
  \hline
  Height &amp; \multicolumn{2}{c|}{Ideal weight}\\
  (cm)   &amp; \multicolumn{2}{c|}{(kg)}\\
  \hline
  155 &amp; 53.5 &amp; 64\\
  \hline
\end{tabular}
```

`r@{--}l` means: right-aligned, then a dash, then left-aligned, with no normal gap between them. The result is "53.5–64". The `@{...}` form removes the usual column space; you can add space back with `\hspace` inside the braces.

CLEANER RULES: BOOKTABS

The primer's tables use `\hline`, but for a cleaner look, the `booktabs` package gives `\toprule`, `\midrule` and `\bottomrule`, with no vertical lines. Use `\cmidrule{2-3}` for a partial rule. This is an extension beyond the primer, but it is standard practice.

7.3 Better columns with the array package

```
\usepackage{array}
\begin{tabular}{|m{1.5cm}|m{2.3cm}|}
\hline
Planet &amp; Mean distance from sun (km)\\
\hline
Mercury &amp; 58100000\\
\hline
\end{tabular}
```

Here the `m` columns are the same width as `p` columns would be, but each cell's text is centred vertically in its box, so a two-line heading sits neatly in the middle.

- `m{width}`: a column of fixed width whose text is centred *vertically*. `p{width}` aligns the text with the top.
- `>{\bfseries}l`: put a declaration in front of every entry of a column. Here, the column is bold.
- `!{...}` is like `@{...}`, but keeps the normal space between columns.

MULTI-ROW CELLS: `MULTIROW`

```
\usepackage{multirow}
\renewcommand{\multirowsetup}{\centering}
\multirow{3}{1.5cm}{Planet}    % spans three rows, vertically centred
```

`\multirow{num}{width}{text}` spans `num` rows. The `\multirowsetup` line centres the text horizontally; without it, the text is left-aligned.

7.4 Tables that run over several pages: longtable

```
\usepackage{longtable}
\begin{longtable}{|c|l|}
  \hline
  Year &amp; Event\\
  \hline
  \endfirsthead          % rows for the first page's header are above this line
  \hline
  Year &amp; Event (continued)\\
  \hline
  \endhead               % header repeated on each following page
  \hline
  continued on the next page\\
  \endfoot               % footer on each page except the last
  \hline
  Source: The Cambridge Factfinder\\
  \endlastfoot          % footer on the last page
  1900 &amp; Max Planck proposes quantum theory\\
  \hline
\end{longtable}
```

Use `longtable` instead of `tabular`, and mark the rows that repeat with `\endfirsthead`, `\endhead`, `\endfoot` and `\endlastfoot`.

OTHER PACKAGES NAMED IN THE PRIMER

`tabularx`, `dcolumn`, `delarray` and `hhline`. Read their manuals with `texdoc tabularx` in a terminal.

Remember (Part 7)

- Use `tabbing` for aligned text, `tabular` for a table.
- Columns are `&`-separated; rows end with `\\`.
- `\hline` and `\cline` draw rules; `|` draws vertical lines.
- `\multicolumn` merges cells. Keep the last column's line in its position argument.
- Use `longtable` for tables longer than a page.